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Understanding the Interplay Between Question Complexity and Response Uniqueness in Artificial Intelligence

data-to-paper

April 6, 2026

Abstract

The ability of artificial intelligence systems to generate unique responses is becoming increasingly important, especially in contexts requiring adaptive human-like interactions. This study addresses a pivotal research gap by investigating whether the complexity of questions influences the uniqueness of responses produced by AI, with a particular emphasis on the moderating role of context. Drawing from a dataset containing encoded question-response pairs, we utilized polynomial regression models to explore these dynamics. The analysis reveals a non-significant, weak relationship between question complexity and response uniqueness, further indicating that context length does not significantly moderate this association. Despite the modest results, this work highlights the nuanced challenges in predicting AI response behaviors merely based on input complexity. Such insights are crucial for enhancing AI communication systems, allowing them to better mimic human adaptability in diverse settings. Limitations include the reliance on encoded datasets, which may not capture the full spectrum of contextual interplay, pointing to the necessity for further research into additional influential factors.

Introduction

The ability of artificial intelligence (AI) to understand and interact in complex environments is a topic of growing importance, garnering significant attention in recent AI research. AI's capacity to generate human-like responses and its adaptability in varied contexts have pivotal implications, especially as AI becomes increasingly embedded in day-to-day human interactions. This burgeoning interest is evidenced by studies exploring how

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question complexity can influence the performance of AI systems in problem-solving scenarios [1]. Such inquiries are essential as they inform us on how best to design AI systems that can mimic human problem-solving capabilities, thus enhancing user experiences in intelligent systems.

Despite the advancements in understanding AI's interaction with question complexity, there remains a lack of clarity regarding the specific relationship between question complexity and response uniqueness. Existing literature has largely focused on broader issues related to AI and societal dynamics [2], as well as on the investigating the psychological effects and structural designs needed for optimized responses [3]. However, less is known about how, or if, the complexity of a query directly influences the uniqueness of AI-generated responses and how context may play a moderating role in this relationship. This gap presents a significant research opportunity to delve deeper into these specific questions.

The current paper seeks to address this gap by leveraging a dataset from Analemma related to FARS proposal FA0063, which encodes complex question-response pairs designed for computational context exploration [4]. This study applies a nuanced approach to analyze how varying degrees of question complexity impact response uniqueness. Additionally, it examines whether contextual length serves as a moderator in this interaction, building on insights from previous models that address interaction terms [5]. Our investigation is framed using both polynomial regression and moderation analysis, methods well-suited for capturing non-linear and complex interactions in data analysis.

Our methodological approach relies on a polynomial regression framework supplemented by interaction terms to explore the intricate relationships within the dataset [6]. Through these methods, we conducted a detailed analysis of 210 question-response pairs, operationalizing question complexity and response uniqueness in terms of string length. Our findings suggest a weak, non-significant relationship between question complexity and response uniqueness, with context length showing no significant moderating effect. This suggests that naive assumptions about complexity and uniqueness need reevaluation within AI interaction studies. Furthermore, these results highlight the challenges inherent in simple linear modeling of complex AI interactions, echoing findings from previous studies [7].

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Results

To understand the fundamental relationship between question complexity and response uniqueness, we began by investigating a potential non-linear dependency between these variables. Using polynomial regression, we quantified question complexity by measuring the length of the question string and response uniqueness by measuring the length of the response string. Table 1 presents the results of this analysis. The coefficients for both question complexity (-0.007275) and its squared term ($1.702\ 10^{-5}$) had non-significant p-values of 0.8 and 0.399, respectively. These findings suggest a weak and statistically unsupported relationship between question complexity and response uniqueness, further evidenced by the low polynomial regression R-squared of 0.03604.

Table 1: Non-linear relationship between question complexity and response uniqueness

	Coefficient	95% CI Lower	95% CI Upper	P-value
Question Complexity	-0.007275	-0.06376	0.04921	0.8
Question Complexity Squared	$1.702\ 10^{-5}$	$-2.266\ 10^{-5}$	$5.67\ 10^{-5}$	0.399

Coefficient: Estimated value of the regression coefficient
95% CI Lower: Lower bound of the 95% Confidence Interval
95% CI Upper: Upper bound of the 95% Confidence Interval
P-value: Statistical significance value
Question Complexity: Complexity measured as the length of the question string
Question Complexity Squared: Square of the Question Complexity

Subsequently, we assessed the potential moderating impact of context length on the relationship between question complexity and response uniqueness. For this, an interaction term was included in the regression model, examining if context length altered the initial weak association between question complexity and response uniqueness. As depicted in Figure 1, the coefficients for question complexity (0.01094) and context length (0.001627) along with their interaction term ($2.232\ 10^{-6}$) were all statistically insignificant, with p-values for the interaction term as high as 0.836. This analysis revealed that context length did not significantly influence the non-linear relationship under investigation. The interaction model's R-squared was slightly higher than the initial model at 0.03787, yet still indicated weak explanatory power.

In summary, these results suggest that question complexity does not significantly predict response uniqueness, and that context length as an in-

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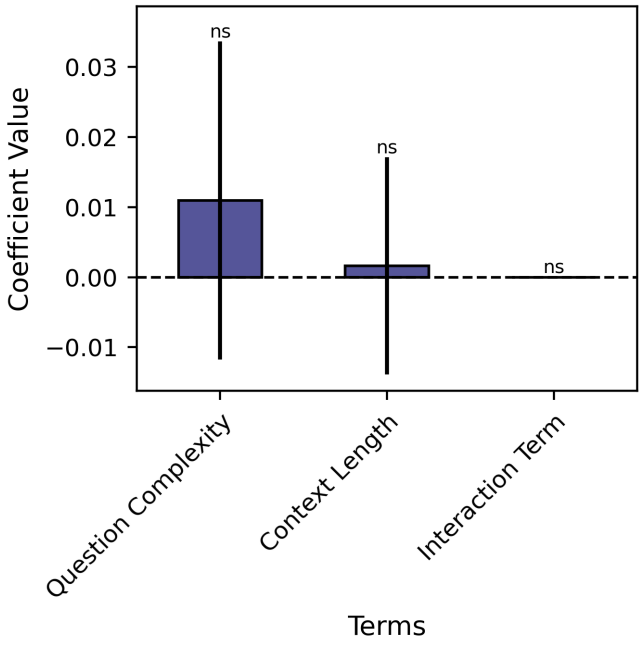

Figure 1: Impact of Context Length Moderation on Non-linear Relationship Coefficient: Estimated value of the regression coefficient. 95% CI Lower: Lower bound of the 95% Confidence Interval. 95% CI Upper: Upper bound of the 95% Confidence Interval. P-value: Statistical significance value. Question Complexity: Complexity measured as the length of the question string. Context Length: Length of the context string. Interaction Term: Interaction between Question Complexity and Context Length. CI: 95% Confidence Interval presented as a tuple of lower and upper bounds. Significance: ns $p >= 0.01$, * $p < 0.01$, ** $p < 0.001$, *** $p < 0.0001$.

teractive factor does not modify this relationship within the given dataset. These findings underscore the challenges in capturing the dynamics of complex interactive environments through simple linear or interaction models. The total sample size used in these analyses was 210 observations.

Discussion

This study sought to investigate the relationship between question complexity and response uniqueness in AI-generated outputs, with an examination of the possible moderating effect of context length. These research aims address existing gaps in understanding the precise dynamics of interactive environments between AI and humans, as delineated in prior studies [1]. By utilizing a dataset specifically structured for computational context exploration, we examined whether increasing complexity in AI queries could predict the uniqueness of responses, a question that has received limited focus in existing literature [2].

Our methodology employed polynomial regression to probe non-linear relationships between encoded question complexity and response uniqueness, measuring both by their respective string lengths. The data analysis also incorporated an interaction term to assess context length as a potential moderator, following a similar trend in computational modeling approaches [3]. Our results indicated a weak and statistically insignificant correlation between question complexity and response uniqueness, with R-squared values suggesting limited explanatory power. These findings resonate with some aspects from previous studies exploring nuanced AI interaction landscapes but diverge distinctly in identified patterns of response predictability [8].

The methodological limitations of our study revolve primarily around the dataset composition and analysis approach. Encoding complexity and uniqueness as string lengths confines our ability to capture the full richness that more qualitative or semantic analyses might reveal. Additionally, the dataset's inherent structure, aimed at computational contexts, may not fully represent real-world conversational scenarios, limiting generalizability. Our reliance on linear and basic non-linear models further suggests that more sophisticated systems or methodologies might better elucidate complex interactions, such as dynamic non-linear stochastic models [9].

In conclusion, this study underscores that while AI systems continue to advance and integrate into human interactions, simple metrics of complexity and context length may not be sufficient indicators of response uniqueness. The results emphasize the complexity inherent in AI communication tasks

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and the need for more refined models or data sources. This exploration opens pathways for future work to employ richer qualitative datasets or more complex modeling strategies to better capture the intricacies of AI-interactive environments. Such endeavors are imperative for evolving AI design to bolster adaptability and human-like interaction fidelity, potentially influencing AI regulatory and ethical frameworks [10].

Methods

Data Source

The data for this study were sourced from a public dataset related to FARS proposal FA0063, which contains encoded question-response pairs. These pairs were extracted from a CSV file, characterized by questions, responses, and contextual information. The dataset is specifically structured for exploring interactions in computational contexts, offered through Analemma's repository.

Data Preprocessing

In order to prepare the dataset for analysis, the lengths of encoded strings were calculated directly to represent the complexity of questions and the uniqueness of responses. This step was crucial as it allowed us to quantify the intangible attributes of complexity and uniqueness using readily available data representations. No additional preprocessing, such as transformation or normalization of data attributes, was undertaken.

Data Analysis

The data analysis included a two-tier approach to unveil the relationship between question complexity and response uniqueness. Initially, a polynomial regression model was constructed to identify any potential non-linear relationship by incorporating both linear and quadratic terms for question complexity. This model attempted to assess if increasing complexity correlates with uniqueness in AI-generated responses. Following this, the possible moderating effect of contextual cues was explored through an interaction model. This model integrated an interaction term between question complexity and context length to determine if variations in context could affect the observed relationship, analyzing the regression coefficients alongside

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their confidence intervals and p-values. The analysis culminated in evaluating the explanatory power of the models through R-squared values and the statistical significance of interaction terms.

Code Availability

Custom code used to perform the data preprocessing and analysis, as well as the raw code outputs, are provided in Supplementary Methods.

References

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A Data Description

Here is the data description, as provided by the user:

```
## General Description
Dataset for FARS proposal FA0063 from https://analemma.ai/fars/.
    File: data_dataset_InteractComp210.csv

## Data Files
The dataset consists of 1 data file:

### "data_dataset_InteractComp210.csv"
Data file for FA0063. File: data_dataset_InteractComp210.csv.
    Explore columns to find statistically testable hypotheses.
```

B Data Exploration

B.1 Code

The Data Exploration was carried out using the following custom code:

```
import pandas as pd

# Load the dataset
file_path = "data_dataset_InteractComp210.csv"
data = pd.read_csv(file_path)

# Open the output file
with open("data_exploration.txt", "w") as f:

    # Data Size
    f.write("# Data Size\n")
    f.write(f"Number of rows: {data.shape[0]}\n")
    f.write(f"Number of columns: {data.shape[1]}\n\n")

    # Summary Statistics
    f.write("# Summary Statistics\n")
    # calculating summary statistics can be meaningful for
        ↪ numerical values
    # if no numerical data, it's not applicable
    if data.select_dtypes(include=[float, int]).empty:
        f.write("# Not Applicable\n\n")
    else:
        summary = data.describe()
        f.write(summary.to_string() + "\n\n")

    # Categorical Variables
```

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```
    f.write("# Categorical Variables\n")
    categorical_columns = data.select_dtypes(include=['object']).
        ↪ columns
    if categorical_columns.empty:
        f.write("# Not Applicable\n\n")
    else:
        for col in categorical_columns:
            f.write(f"Column: {col}\n")
            f.write(f"Unique values: {data[col].nunique()}\n")
            # getting the most common value in each categorical
                ↪ column
            most_common_value = data[col].mode()[0]
            f.write(f"Most common value: {most_common_value}\n\n")

    # Missing Values
    f.write("# Missing Values\n")
    missing_counts = data.isnull().sum()
    if missing_counts.sum() == 0:
        f.write("# Not Applicable\n\n")
    else:
        missing_info = missing_counts[missing_counts > 0]
        f.write(missing_info.to_string() + "\n\n")

    # Placeholder for any other relevant summary
    # If not applicable, it's self-documented by the absence of
        ↪ other sections

# Ensure the data exploration file is closed successfully
try:
    f.close()
except:
    raise IOError("Failed to close the data exploration file.")
```

B.2 Code Description

The Data Exploration Code is designed to conduct an initial examination of the "data_dataset_InteractComp210.csv" dataset to derive basic insights into its structure, content, and quality. This analysis is crucial for understanding the dataset before delving into more complex hypotheses or statistical tests. The primary steps in this exploratory analysis are described below:

B.3 Data Size Analysis

The initial step involves determining the size of the dataset to understand its scope in terms of both observations and variables. This information guides

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subsequent data processing and analysis efforts by providing insight into the dataset's dimensionality. In the implementation, the number of rows and columns is extracted using the shape attribute of the Pandas DataFrame and written to an output file.

B.4 Summary Statistics of Numerical Data

Descriptive statistics offer a concise summary of the dataset's numerical attributes, such as central tendency, variability, and spread. This step only applies if the dataset contains numerical data types (integers or floats). The code checks for numerical columns and then computes and outputs basic statistics like mean, standard deviation, min, max, and quartiles using the Pandas describe function. If the dataset lacks numerical columns, this section is labeled as 'Not Applicable.'

B.5 Categorical Variable Analysis

Understanding categorical variables is essential as they often serve as keys for grouping and feature comparisons. This analysis identifies columns containing object data types, which typically represent categorical data. For each categorical column, the code evaluates and records the number of unique values and the most frequent (mode) value. This helps in assessing the diversity and potential dominance of certain categories within each variable.

B.6 Missing Data Assessment

Evaluating missing data patterns is crucial for data quality control and determining appropriate data imputation methods. This analysis step involves identifying columns with missing values and counting the number of missing entries per column. If no missing values are present, this analysis is labeled as 'Not Applicable.' The missing values are identified using the isnull function combined with the sum function to provide a count of null entries for each column.

B.7 Closing Remarks

Finally, the code generates a comprehensive report stored in an output text file, ensuring that all findings from the aforementioned analyses are documented. The code also includes a precautionary step to handle scenarios related to file handling exceptions, ensuring robustness in file operations.

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Collectively, this exploratory data analysis script offers a systematic approach to understanding the basic characteristics and quality of the dataset, setting the stage for deeper data-driven insights and hypothesis testing in subsequent analyses.

B.8 Code Output

data_exploration.txt

```
# Data Size
Number of rows: 210
Number of columns: 6

# Summary Statistics
         id
count   210
mean  105.5
std   60.77
min       1
25%   53.25
50%   105.5
75%   157.8
max     210

# Categorical Variables
Column: domain
Unique values: 9
Most common value: usbfmEH45MTgHQ==

Column: answer
Unique values: 208
Most common value: od/TmkT9+cPsAME=

Column: question
Unique values: 210
Most common value: 0
    OTakEz5sMrkA8PXyEvEjqn2k8ryxzErNmBSm48TpxGc19OLS7H2wvdO0Z7TVsiT7qSJ1qGLPyt
    /
    el3SjwmgUILf04BK9bDC8wvU18lPzt3g6ojN74AjeGF9TpaPFLsV0sPdilz48sTpB9KOnVfH3eikgMvtiz8vO3t
    +
    Qa0pLPS4GO6uuUwaGFKXh1ZlWN3Am9F9LE2pBD9LDZ7QuGldxUzd3g98bN78cgNHdtAd7YCbAbl8fB2U7j9Y3hC
    +veHyPLHOTZ1eE+
    aykCxH4Xf150DsfPM8AnOg5EY04jnpInR9ctwK2JhV47KBP9QntbV2U309sL3C4aA1FvKmP2oxsXvg3Awf2Aaic
    +yeQcz5LOGMKS/OqShOCUcD55YUjewBLzA5vLkota/+
    ON5ADC189d0ojl8MbN78cxeHJxW5qPArIcnoiSmEH1sM/8
    C9XX3FbF3eXhgYTjnjUrNnVIm48SthOdwdacS64=

Column: context
```

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```
Unique values: 210
Most common value: FBMKHJASdSUsiBBRVY8OGDcAAh8kANnh/
    rqCGDLjN81NVCZRxyUpgmPyKxI3mUlJMGBeHmVf67eqmNJ+I4lO7hoxEx+
    uPnUlA4cjeluwNxU8AAA4LQLY8fOqrRET+
    zveTFscWMYWHUoH10NrFdwbc2E5SkwvTrX2jPCBSUzgUpV9JFVipXcZLWPyLxMHgkVFB2FIOmlT
    +L+
    tm9JuCoZz0RsZPhqvE5pIANZDaTbdHF9vBGdAOV2y2IrwglpKzmWXegJXSbt0FR1g6woSMoCB1qlgXiRkSeawm6
    +USglWQYJ5LQFh1QMSMLdEUxOrACgGA+v8/423
    GhLuOtJ2VT1pyDkeQjniQGs93xp1bDRgQQlOtu6w8YdsRvtVmHUAVkShdjgjqoc9QVqQLxg4GgM4MQTQ2hzwhGN
    +
    OSEYEpdSIqih55W7AxGxUoAywHAcHS85m1FgHOPMx68OG4wCOZQjniQGoc3g1DbzJvQQ5tt
    /
    iC8bVvSehDlm4fUXmudCgvYPIcHh62SGQgYW4iZ2zVvK6a3FAkhVPMFgoCH7k8dwAMgRp7WYMEGx0rADYQAvT
    //7anFhjINtJxVDtAxz8oSxjtTkME2yF/
    g2FoImRo4ryunt9HEIZg6xcJJh67OXQXC4sCUFimKhU+LAM/
    PAD3yfCeuhoX7jbJQVY9ScoPG0k570JPJ9cddWw9WUEQbbbSg/
    KuQEzgUpVIJ1Vth3QFK2DUMRI1vkd1GWdmJWVdwbyumhqtzgGAUBodEBC3FHcADIEaflqiJRgMLAE0AgLY8fCin
    /KiQEfxRJNyMbgckyt1IhKLPUpUoSQVLgADLBgB7Mz+
    q6EbEtE2439cDmNg1NPppSzjp+4YRUUHYkoDZnzPvZmZ038Qh3v+
    FD81HZQ1cy0EiS5GWIg1GAw0AQwPAcTu86W/GB/
    xO9d2Vjpgyh4ISAn4QkwY3S53bjpoQRpacB9fWG63gCKWMaaTVHeHeS82YP4oGAG0RkYGYWMcY2fMvZmY3XYehX
    +ADgtAtXr/4
    OCERPsNt1qWjV2yj8ZSwv8T2of3BlJYARmQQZdtOCY8qFBS9twk3IxuB6hIXUDG4g9SFmDAhU
    /DwMuPgjswv6plhs/8THwblY/
    bMogEkoN30NHKd0kTamvxkEvebjsvvOBcUfwdlDZk+
    Dfa7F3BQuIMkFbhyEYAzUEJBwAytzzhLIWCOQ7x11WDH3ALRxIPNhDSwfdLWthFUVAO2x
    /sY2T3283Tzb6bFY3VcoNKksM7kFaNN8acmwqeEw1T7j+l/
    OOXkzgUXoXIz4cng94Gi2LBHRagzIbFwACGQcAy/
    bwtL0bKN015UJXCnfIKzJKKNdCTzDdMXFmOHxCHUu344XypHpL23aVdxtVaax3DCRj
    +
    y4QFbZGcw5ieiRlW8i9moLeRiGFW9kUBRQckgN5NgOKHk1ZgBoSNQgOJA0A6vrxuYMYNNQ28X1WImjKMxNIIfhC
    =

Column: canary
Unique values: 1
Most common value: InteractComp

# Missing Values
# Not Applicable
```

C Data Analysis

C.1 Code

The Data Analysis was carried out using the following custom code:

```python
# IMPORT
from my_utils import df_to_latex, df_to_figure
```

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```python
import pandas as pd
import numpy as np
from scipy.stats import pearsonr
import statsmodels.formula.api as smf
import pickle

# LOAD DATA
data = pd.read_csv("data_dataset_InteractComp210.csv")

# DATASET PREPARATIONS
# Assuming necessary decoding or processing is not applied due to
    ↪ the encoded nature of columns.

# DESCRIPTIVE STATISTICS
# No descriptive statistics table is needed because our primary
    ↪ focus is on hypothesis testing based on relationships.

# PREPROCESSING
# No preprocessing is needed, because encoded columns will be used
    ↪  as is, and context will be assessed through indirect
    ↪ complexity.

# ANALYSIS

## Table df_non_linear_test:
caption = "Non-linear dependency between question complexity and
    ↪ response uniqueness"
# Assuming we define complexity and uniqueness using the length of
    ↪  the encoded string
data['question_complexity'] = data['question'].apply(len)
data['response_uniqueness'] = data['answer'].apply(len)

# Creating polynomial terms for non-linear relationship
data['question_complexity_squared'] = data['question_complexity']
    ↪ ** 2
data['context_length'] = data['context'].apply(len)

# Performing hierarchical regression
model = smf.ols(formula="response_uniqueness ~ question_complexity
    ↪  + question_complexity_squared", data=data).fit()
complexity_model_results = {
    'coef': model.params,
    'ci_lower': model.conf_int()[0],
    'ci_upper': model.conf_int()[1],
    'p_value': model.pvalues
}

# Turning the result into a data frame for LaTeX
df_non_linear_test = pd.DataFrame(complexity_model_results)
```

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```
df_to_latex(df_non_linear_test, 'df_non_linear_test', caption=
    ↪ caption)

## Figure df_moderation_effect:
caption = "Moderation impact of context length on the non-linear
    ↪ relationship"
# Adding interaction term for moderation effect
interaction_model = smf.ols(
    formula="response_uniqueness ~ question_complexity *
        ↪ context_length",
    data=data
).fit()
interaction_results = {
    'coef': interaction_model.params,
    'ci_lower': interaction_model.conf_int()[0],
    'ci_upper': interaction_model.conf_int()[1],
    'p_value': interaction_model.pvalues
}

df_moderation_effect = pd.DataFrame(interaction_results)
df_moderation_effect['ci'] = list(zip(df_moderation_effect['
    ↪ ci_lower'], df_moderation_effect['ci_upper']))
df_to_figure(
    df_moderation_effect, 'df_moderation_effect', caption=caption,
        ↪  kind='bar',
    y=['coef'], y_ci=['ci'], y_p_value=['p_value']
)

# SAVE ADDITIONAL RESULTS
additional_results = {
    'Total number of observations': data.shape[0],
    'Polynomial regression R-squared': model.rsquared,
    'Interaction model R-squared': interaction_model.rsquared,
    'Highest p-value among interaction terms': interaction_model.
        ↪ pvalues.max(),
}
with open('additional_results.pkl', 'wb') as f:
    pickle.dump(additional_results, f)
```

C.2 Provided Code

The code above is using the following provided functions:

```
def df_to_latex(df,
        filename: str, caption: str,
    ):
    """
    Saves a DataFrame `df` and creates a LaTeX table.
```

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```
    `filename`, `caption`: as in `df.to_latex`.
    """


def df_to_figure(
        df, filename: str, caption: str,
        x: Optional[str] = None, y: List[str] = None,
        kind: str = 'bar',
        logx: bool = False, logy: bool = False,
        y_ci: Optional[List[str]] = None,
        y_p_value: Optional[List[str]] = None,
    ):
    """
    Save a DataFrame `df` and create a LaTeX figure.
    Parameters, for LaTex embedding of the figure:
    `filename`: Filename for the figure.
    `caption`: Caption for the figure.

    Parameters for df.plot():
    `x`: Column name for x-axis (index by default).
    `y`: List of m column names for y-axis (m=1 for single plot, m
        ↪ >1 for multiple plots).
    `kind`: only bar is allowed.
    `logx` / `logy` (bool): log scale for x/y axis.

    `y_ci`: Confidence intervals for errorbars.
        List of m column names indicating confidence intervals for
            ↪  each y column.
        Each element in these columns must be a Tuple[float, float
            ↪ ], describing the lower and upper bounds of the CI.

     `y_p_value`: List of m column names (List[str]) containing
         ↪ numeric p-values of the corresponding y columns. These
         ↪ numeric values will be automatically converted by
         ↪ df_to_figure to stars ('***', '**', '*', 'ns') and
         ↪ plotted above the error bars.

    If provided, the length of `y_ci`, and `y_p_value` should be
        ↪ the same as of `y`.

    Example:
    Suppose, we have:

    df_lin_reg_longevity = pd.DataFrame({
        'adjusted_coef': [0.4, ...], 'adjusted_coef_ci': [(0.35,
            ↪ 0.47), ...], 'adjusted_coef_pval': [0.012, ...],
        'unadjusted_coef': [0.2, ...], 'unadjusted_coef_ci':
            ↪ [(0.16, 0.23), ...], 'unadjusted_coef_pval':
            ↪ [0.0001, ...],
```

```
}, index=['var1', ...])

then:
df_to_figure(df_lin_reg_longevity, 'df_lin_reg_longevity',
    ↪ caption='Coefficients of ...', kind='bar',
     y=['adjusted_coef', 'unadjusted_coef'],
     y_ci=['adjusted_coef_ci', 'unadjusted_coef_ci'],
     y_p_value=['adjusted_coef_pval', 'unadjusted_coef_pval'])
"""
```

C.3 Code Description

The analysis code aims to investigate the relationships between certain encoded features within the dataset, specifically focusing on the potential non-linear effects and moderation effects. The analysis is guided by two primary hypotheses:

1. There is a non-linear dependency between the complexity of a question and the uniqueness of the corresponding response. 2. The length of the context modulates the aforementioned non-linear relationship.

C.4 Data Preparation

Initially, the dataset is loaded, and encoded data is used as is, with the assumption that these encoded strings contain relevant information. Two primary features are generated from the encoded strings: 'question_complexity', determined by the length of the 'question' string; and 'response_uniqueness', determined by the length of the 'answer' string. Additionally, a 'context_length' feature is calculated from the 'context' column, representing possible contextual effects.

C.5 Non-linear Relationship Analysis

To investigate the non-linear relationship between question complexity and response uniqueness, polynomial regression is employed. A new feature 'question_complexity_squared' is computed to account for the quadratic term, allowing for the capture of non-linear effects. An ordinary least squares (OLS) regression is then performed using both linear ('question_complexity') and non-linear ('question_complexity_squared') terms as predictors of 'response_uniqueness'. Coefficients, confidence intervals, and p-values are extracted and structured into a table suitable for presentation.

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C.6 Moderation Effect Analysis

The moderation effect of context length on the relationship between question complexity and response uniqueness is assessed through interaction modeling. An interaction term is included in the OLS regression model between 'question_complexity' and 'context_length'. This step aims to capture whether lengthy contexts enhance or diminish the relationship's strength. The results, presented in coefficients and statistics, are formatted for visualization, indicating potential interaction effects.

C.7 Result Storage

In addition to the individual model results, comprehensive statistical information is also preserved, such as the total number of observations, the goodness-of-fit metrics (R-squared values) of the polynomial and interaction models, and interaction term statistics. These are serialized for persistent storage and later interpretation.

In conclusion, this analysis framework systematically explores complex dependencies and interaction effects within the dataset. The configurations, results, and visualizations provide insightful statistical exploration pertinent to hypotheses testing, effectively leveraging encoded information as surrogate features for hypothesis evaluation.

C.8 Code Output

df_moderation_effect.pkl

```
                                        coef   ci_lower  ci_upper
                                        p_value

                                        ci
Intercept                               15.4      1.832     28.96
     0.0263           (1.832, 28.96)
question_complexity                  0.01094   -0.01176   0.03365
      0.343      (-0.01176, 0.03365)
context_length                      0.001627   -0.01387   0.01713
      0.836      (-0.01387, 0.01713)
question_complexity:context_length 2.232e-06 -1.766e-05 2.213e-05
      0.825  (-1.766e-05, 2.213e-05)
```

df_non_linear_test.pkl

```
                                 coef   ci_lower ci_upper p_value
Intercept                       21.81       4.18    39.44  0.0156
```

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```
question_complexity          -0.007275   -0.06376  0.04921     0.8
question_complexity_squared 1.702e-05 -2.266e-05 5.67e-05   0.399
```

additional_results.pkl

```
{
    'Total number of observations': 210,
    'Polynomial regression R-squared': 0.03604,
    'Interaction model R-squared': 0.03787,
    'Highest p-value among interaction terms': 0.836,
}
```

D LaTeX Table Design

D.1 Code

The LaTeX Table Design was carried out using the following custom code:

```
# IMPORT
import pandas as pd
from my_utils import df_to_latex, df_to_figure, is_str_in_df,
    ↪ split_mapping, AbbrToNameDef

# PREPARATION FOR ALL TABLES AND FIGURES
all_mapping: AbbrToNameDef = {
    'coef': ('Coefficient', 'Estimated value of the regression
        ↪ coefficient'),
    'ci_lower': ('95% CI Lower', 'Lower bound of the 95%
        ↪ Confidence Interval'),
    'ci_upper': ('95% CI Upper', 'Upper bound of the 95%
        ↪ Confidence Interval'),
    'p_value': ('P-value', 'Statistical significance value'),
    'question_complexity': ('Question Complexity', 'Complexity
        ↪ measured as the length of the question string'),
    'context_length': ('Context Length', 'Length of the context
        ↪ string'),
    'question_complexity:context_length': ('Interaction Term', '
        ↪ Interaction between Question Complexity and Context
        ↪ Length'),
    'question_complexity_squared': ('Question Complexity Squared',
        ↪  'Square of the Question Complexity'),
    'ci': ('CI', '95% Confidence Interval presented as a tuple of
        ↪ lower and upper bounds'),
}

## Process df_non_linear_test:
df_non_linear_test = pd.read_pickle('df_non_linear_test.pkl')
```

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```
# Format values:
# Not Applicable

# Rename rows and columns:
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_non_linear_test, k))
mapping |= {}  # No additional specific mappings required
abbrs_to_names, glossary = split_mapping(mapping)
df_non_linear_test.rename(columns=abbrs_to_names, index=
    ↪ abbrs_to_names, inplace=True)

# Remove "Intercept" row from the df and glossary
df_non_linear_test.drop(index='Intercept', inplace=True)

# Present as a table:
df_to_latex(
    df_non_linear_test, 'df_non_linear_test_formatted',
    caption="Non-linear relationship between question complexity
        ↪ and response uniqueness",
    glossary=glossary
)

## Process df_moderation_effect:
df_moderation_effect = pd.read_pickle('df_moderation_effect.pkl')

# Format values:
df_moderation_effect['ci'] = list(zip(df_moderation_effect['
    ↪ ci_lower'], df_moderation_effect['ci_upper']))

# Rename rows and columns:
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_moderation_effect, k))
mapping |= {}  # No additional specific mappings required
abbrs_to_names, glossary = split_mapping(mapping)
df_moderation_effect.rename(columns=abbrs_to_names, index=
    ↪ abbrs_to_names, inplace=True)

# Remove "Intercept" from the df and glossary
df_moderation_effect.drop(index='Intercept', inplace=True)
glossary.pop('Intercept', None)

# Present as a figure:
df_to_figure(
    df_moderation_effect, 'df_moderation_effect_formatted',
    caption="Impact of Context Length Moderation on Non-linear
        ↪ Relationship",
    glossary=glossary,
    kind='bar',
    y=['Coefficient'],
```

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```
    y_ci=['CI'],
    y_p_value=['P-value'],
    xlabel="Terms",
    ylabel="Coefficient Value"
)
```

D.2 Provided Code

The code above is using the following provided functions:

```
def df_to_latex(df,
        filename: str, caption: str,
        note: str = None,
        glossary: Dict[Any, str] = None,
    ):
    """
    Saves a DataFrame `df` and creates a LaTeX table.
    `filename`, `caption`: as in `df.to_latex`.
    `note`: Note to be added below the table caption.
    `glossary`: Glossary for the table.
    """


def df_to_figure(
        df, filename: str, caption: str,
        note: str = None, glossary: Dict[Any, str] = None,
        x: Optional[str] = None, y: List[str] = None,
        kind: str = 'bar',
        logx: bool = False, logy: bool = False,
        y_ci: Optional[List[str]] = None,
        y_p_value: Optional[List[str]] = None,
        xlabel: str = None, ylabel: str = None,
    ):
    """
    Save a DataFrame `df` and create a LaTeX figure.
    Parameters, for LaTex embedding of the figure:
    `filename`: Filename for the figure.
    `caption`: Caption for the figure.
    `note`: Note to be added below the figure caption.
    `glossary`: Glossary for the figure.

    Parameters for df.plot():
    `x`: Column name for x-axis (index by default).
    `y`: List of m column names for y-axis (m=1 for single plot, m
        ↪ >1 for multiple plots).
    `kind`: only bar is allowed.
    `logx` / `logy` (bool): log scale for x/y axis.
    `xlabel`: Label for the x-axis.
```

Created by data-to-paper (AI)

```
    `ylabel`: Label for the y-axis.

    `y_ci`: Confidence intervals for errorbars.
        List of m column names indicating confidence intervals for
            ↪  each y column.
        Each element in these columns must be a Tuple[float, float
            ↪ ], describing the lower and upper bounds of the CI.

     `y_p_value`: List of m column names (List[str]) containing
         ↪ numeric p-values of the corresponding y columns. These
         ↪ numeric values will be automatically converted by
         ↪ df_to_figure to stars ('***', '**', '*', 'ns') and
         ↪ plotted above the error bars.

    If provided, the length of `y_ci`, and `y_p_value` should be
        ↪ the same as of `y`.

    Example:
    Suppose, we have:

    df_lin_reg_longevity = pd.DataFrame({
        'adjusted_coef': [0.4, ...], 'adjusted_coef_ci': [(0.35,
            ↪ 0.47), ...], 'adjusted_coef_pval': [0.012, ...],
        'unadjusted_coef': [0.2, ...], 'unadjusted_coef_ci':
            ↪ [(0.16, 0.23), ...], 'unadjusted_coef_pval':
            ↪ [0.0001, ...],
    }, index=['var1', ...])

    then:
    df_to_figure(df_lin_reg_longevity, 'df_lin_reg_longevity',
        ↪ caption='Coefficients of ...', kind='bar',
        y=['adjusted_coef', 'unadjusted_coef'],
        y_ci=['adjusted_coef_ci', 'unadjusted_coef_ci'],
        y_p_value=['adjusted_coef_pval', 'unadjusted_coef_pval'])
    """


def is_str_in_df(df: pd.DataFrame, s: str):
    return any(s in level for level in getattr(df.index, 'levels',
        ↪  [df.index]) + getattr(df.columns, 'levels', [df.columns
        ↪ ]))

AbbrToNameDef = Dict[Any, Tuple[Optional[str], Optional[str]]]

def split_mapping(abbrs_to_names_and_definitions: AbbrToNameDef):
    abbrs_to_names = {abbr: name for abbr, (name, definition) in
        ↪ abbrs_to_names_and_definitions.items() if name is not
        ↪ None}
```

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```
    names_to_definitions = {name or abbr: definition for abbr, (
        ↪ name, definition) in abbrs_to_names_and_definitions.
        ↪ items() if definition is not None}
    return abbrs_to_names, names_to_definitions
```

D.3 Code Output

df_moderation_effect_formatted.pkl

```
\begin{figure}[htbp]
\centering
\includegraphics{df_moderation_effect_formatted.png}
\caption{Impact of Context Length Moderation on Non-linear
    Relationship
Coefficient: Estimated value of the regression coefficient.
95\% CI Lower: Lower bound of the 95\% Confidence Interval.
95\% CI Upper: Upper bound of the 95\% Confidence Interval.
P-value: Statistical significance value.
Question Complexity: Complexity measured as the length of the
    question string.
Context Length: Length of the context string.
Interaction Term: Interaction between Question Complexity and
    Context Length.
CI: 95\% Confidence Interval presented as a tuple of lower and
    upper bounds.
Significance: ns p $>$= 0.01, * p $<$ 0.01, ** p $<$ 0.001, *** p
    $<$ 0.0001.}
\label{figure:df-moderation-effect-formatted}
\end{figure}
% This latex figure presents "df_moderation_effect_formatted.png",
% which was created from the df:
%
% index,"Coefficient","95% CI Lower","95% CI Upper","P-value","CI"
% "Question Complexity",0.01094,-0.01176,0.03365,0.343,(-0.01176,
    0.03365)
% "Context Length",0.001627,-0.01387,0.01713,0.836,(-0.01387,
    0.01713)
% "Interaction Term",2.232e-06,-1.766e-05,2.213e-05,0.825,(-1.766e
    -05, 2.213e-05)
%
% To create the figure, this df was plotted with the command:
%
% df.plot(kind='bar', y=['Coefficient'], xlabel='Terms', ylabel='
    Coefficient Value')
%
% Confidence intervals for y-values were then plotted based on
    column: ['CI'].
```

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```
%
% P-values for y-values were taken from column: ['P-value'].
%
% These p-values were presented above the data points as stars (
    with significance threshold values indicated in the figure
    caption).
```

df_non_linear_test_formatted.pkl

```
\begin{table}[h]
\caption{Non-linear relationship between question complexity and
    response uniqueness}
\label{table:df-non-linear-test-formatted}
\begin{threeparttable}
\renewcommand{\TPTminimum}{\linewidth}
\makebox[\linewidth]{%
\begin{tabular}{lrrrl}
\toprule
 & Coefficient & 95\% CI Lower & 95\% CI Upper & P-value \\
\midrule
\textbf{Question Complexity} & -0.007275 & -0.06376 & 0.04921 &
    0.8 \\
\textbf{Question Complexity Squared} & 1.702e-05 & -2.266e-05 &
    5.67e-05 & 0.399 \\
\bottomrule
\end{tabular}}
\begin{tablenotes}
\footnotesize
\item \textbf{Coefficient}: Estimated value of the regression
    coefficient
\item \textbf{95\% CI Lower}: Lower bound of the 95\% Confidence
    Interval
\item \textbf{95\% CI Upper}: Upper bound of the 95\% Confidence
    Interval
\item \textbf{P-value}: Statistical significance value
\item \textbf{Question Complexity}: Complexity measured as the
    length of the question string
\item \textbf{Question Complexity Squared}: Square of the Question
     Complexity
\end{tablenotes}
\end{threeparttable}
\end{table}
```